MATCAP1

KIAA0895L

SCRIPT

Hi, I'm Jas, and the gene I studied was MATCAP1, aka microtubule associated tyrosine carboxypeptidase 1.
Matcap1 is a protein coding gene that creates a protein by the same name. I found that matcap could possibly be able to suppress cancer proliferation through the use of microtubules, which have functions related to cell growth and mitosis, as well as communication between cells.
Surprisingly, kiaa0895l, which is not matcap1's official name according to ncbi, is actually the alias that provided more results while i was researching.
Matcap1 is located in chromosome 16 and regulates microtubule dynamics and function.
It is connected to many genes related to cytoskeleton formation and organization
Thanks to the human protein atlas, matcap1 is associated with pancreatic cancer, where with higher expression of matcap, there is a higher survival probability over a longer period of time.
Therefore, my first hypothesis is that matcap1 plays a role in suppressing pancreatic cancer cells, acting as a tumor suppressor gene.
However, by searching with the ncbi geoprofiles, it was also shown that when leukemia was treated with imatinib mesylate, there were higher expressions of matcap, meaning that it is not only pancreatic cancer that matcap affects. Matcap expression is also lowered when there is pulmonary sarcoidosis, which is inflammation, which could allow for tumor growth, so i can come to a separate conclusion that matcap doesn't only help pancreatic cancer, but suppresses inflammation in some way as a tumor suppresor gene.
However, the fact that matcap expression is lowered when there is doxycline present is a strange fact, as doxycycline also has anti-cancer properties, including anti-inflammation. It feels like it could be an exception, however, if matcap1 was a tsg with similar anti-inflammatory properties, would matcap expression not be higher?
Matcap1 manipulates microtubule functions, which include mitosis, cell migration, and cell signaling and trafficking, which would, if mutated or taken over by cancer cells, be very useful for proliferation. However, microtubule-targeted agents are already used as a form of cancer therapy, meaning matcap would be able to alter the function of microtubules to work against cancer.
The thing still bothering me is that doxycycline's effect on matcap1 was only with mutated p53 in the picture. However, in an environment with no mutated p53 and no cancer, how would doxycycline affect the expression of matcap1? By using the knockout mouse experiment, it could be possible to discover what exactly happens there.
It would also be good to test the function of using matcap1 to alter microtubules (besides just depending on mtas) to see if matcap1 is truly able to suppress cancer, to see if you could alter matcap1s function

GENE

MATCAP1 – microtubule associated tyrosine carboxypeptidase 1

Homo sapiens (human)

Also known as: KIAA0895L

4931428F04Rik shows up a lot in GEO Profiles when searching for KIAA0895L
WHAT IS 4931428F04Rik?

KIAA0895L - Imatinib mesylate effect on EOL-1 eosinophilic leukemia cell line
Annotation: KIAA0895L, KIAA0895 like
Organism: Homo sapiens
Reporter: GPL570, 1558041_a_at (ID_REF), GDS5406, 653319 (Gene ID), AL834156
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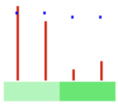

KIAA0895L - Mutant p53 depletion effect on MDA-468.shp53 breast tumor cell line
Annotation: KIAA0895L, KIAA0895 like
Organism: Homo sapiens
Reporter: GPL6244, 8001946 (ID_REF), GDS4096, NM_001040715, BC080183, AY255669, chr16:67209505-67217883 (SPOT ID)
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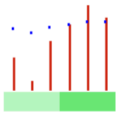

KIAA0895L - H3K9me2 methyltransferase G9a depletion effect on breast cancer cell line
Annotation: KIAA0895L, KIAA0895 like
Organism: Homo sapiens
Reporter: GPL6244, 8001946 (ID_REF), GDS4800, NM_001040715, BC080183, AY255669, chr16:67209505-67217883 (SPOT ID)
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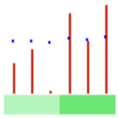

KIAA0895L - Central nervous system primitive neuroectodermal tumors
Annotation: KIAA0895L, KIAA0895 like
Organism: Homo sapiens
Reporter: GPL570, 1558041_a_at (ID_REF), GDS4838, 653319 (Gene ID), AL834156
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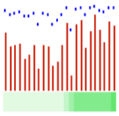

KIAA0895L - Pulmonary sarcoidosis
Annotation: KIAA0895L, KIAA0895 like
Organism: Homo sapiens
Reporter: GPL570, 1558041_a_at (ID_REF), GDS3580, 653319 (Gene ID), AL834156
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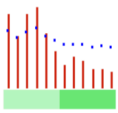
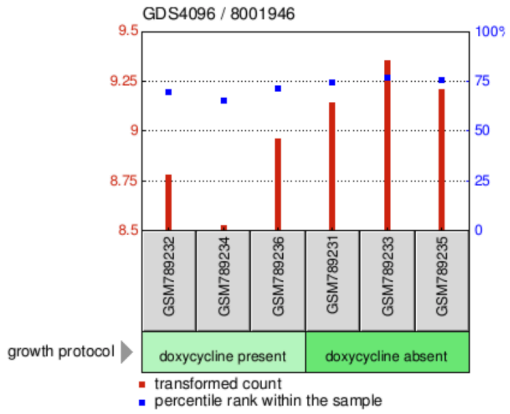


KIAA0895L - Hypoxia effect on monocyte-derived, mature dendritic cells
Annotation: KIAA0895L, KIAA0895 like
Organism: Homo sapiens
Reporter: GPL570, 1558041_a_at (ID_REF), GDS3858, 653319 (Gene ID), AL834156
DataSet type: Expression profiling by array, count, 6 samples
ID: 70383427
GEO DataSets Gene Profile neighbors Chromosome neighbors Homologene neighbors Remove from clipboard

See NCBI clipboard

Profile GDS4096 / 8001946
Title Mutant p53 depletion effect on MDA-468.shp53 breast tumor cell line
Organism Homo sapiens

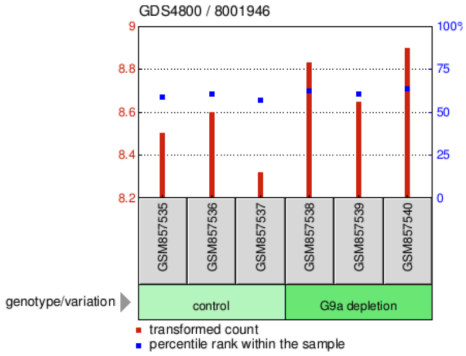


Graph caption help

Sample	Title	Value	Rank
GSM789232	MDA-468.shp53 cells_3D culture, 8 days, +DOX_rep1	8.78368	70
GSM789234	MDA-468.shp53 cells_3D culture, 8 days, +DOX_rep2	8.51219	66
GSM789236	MDA-468.shp53 cells_3D culture, 8 days, +DOX_rep3	8.96595	72
GSM789231	MDA-468.shp53 cells_3D culture, 8 days, -DOX_rep1	9.14894	75
GSM789233	MDA-468.shp53 cells_3D culture, 8 days, -DOX_rep2	9.3579	77
GSM789235	MDA-468.shp53 cells_3D culture, 8 days, -DOX_rep3	9.21299	76

p53: encoded by tp53, is tsg and tsp
doxycycline: anti-inflamatory, etc.

Profile GDS4800 / 8001946
Title H3K9me2 methyltransferase G9a depletion effect on breast cancer cell line
Organism Homo sapiens

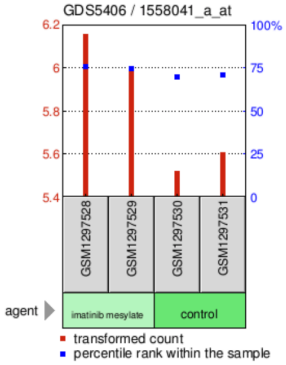


Graph caption help

Sample	Title	Value	Rank
GSM857535	Control cells, rep1	8.50572	59
GSM857536	Control cells, rep2	8.60072	61
GSM857537	Control cells, rep3	8.32232	57
GSM857538	G9a knockdown cells, rep1	8.83357	63
GSM857539	G9a knockdown cells, rep2	8.65279	61
GSM857540	G9a knockdown cells, rep3	8.89982	64

g9a:

Profile GDS5406 / 1558041_a_at
Title Imatinib mesylate effect on EOL-1 eosinophilic leukemia cell line
Organism Homo sapiens

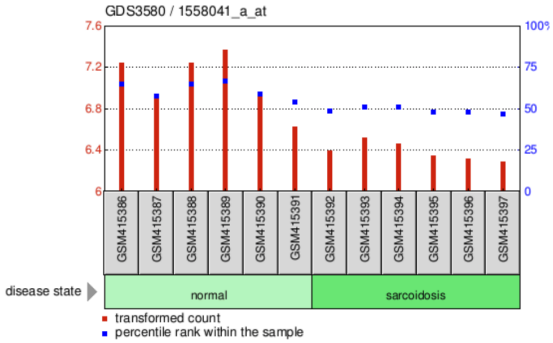


Graph caption help

Sample	Title	Value	Rank
GSM1297528	EOL1_IMA1	6.15869	76
GSM1297529	EOL1_IMA2	6.00562	75
GSM1297530	EOL1_CTRL3	5.52272	70
GSM1297531	EOL1_CTRL4	5.60794	71

Imatinib mesylate:

Profile GDS3580 / 1558041_a_at
Title Pulmonary sarcoidosis
Organism Homo sapiens



Graph caption help

Sample	Title	Value	Rank
GSM415386	Lung-normal-rep1	7.24589	65
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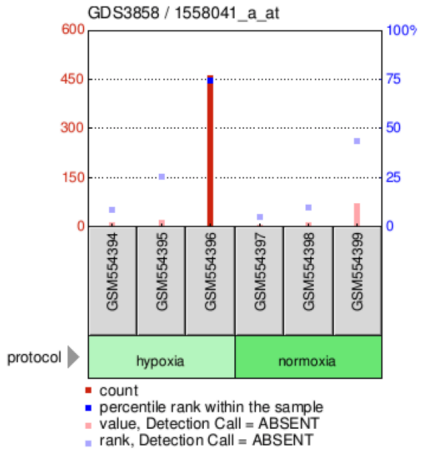
GSM415388	Lung-normal-rep3	7.24483	65
GSM415389	Lung-normal-rep4	7.37189	67
GSM415390	Lung-normal-rep5	6.93959	59
GSM415391	Lung-normal-rep6	6.62945	54
GSM415392	Lung-sarcoidosis-rep1	6.40261	49
GSM415393	Lung-sarcoidosis-rep2	6.52543	51
GSM415394	Lung-sarcoidosis-rep3	6.47223	51
GSM415395	Lung-sarcoidosis-rep4	6.35113	48
GSM415396	Lung-sarcoidosis-rep5	6.32186	48
GSM415397	Lung-sarcoidosis-rep6	6.29361	47

Pulmonary sarcoidosis:

Profile GDS3858 / 1558041_a_at
Title Hypoxia effect on monocyte-derived, mature dendritic cells
Organism Homo sapiens



Graph caption help

Sample	Title	Value	Rank
GSM554394	mDC hypoxia 1	15.7301	9
GSM554395	mDC hypoxia 2	22.925	26
GSM554396	mDC hypoxia 3	464.732	75
GSM554397	mDC normoxia 1	9.26767	5
GSM554398	mDC normoxia 2	6.57637	10
GSM554399	mDC normoxia 3	75.2992	44

Hypoxia:
Nomoxia:
x

GENE LIST (?)

ORIGIN

```
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3421 cctcactgag ctcccagtcc agagagaagc cagacaatta aaacagtaat gacaatga

Multitple transcript variants

MATCAP (https://www.science.org/doi/full/10.1126/science.abn6020)
The genetic screens identified KIAA0895L, an unstudied protein without apparent protease features, as a remaining detyrosinase. Removal of KIAA0895L in combination with VASH1 and VASH2 resulted in undetectable levels of tubulin detyrosination in cultured cells. A crystal structure of KIAA0895L revealed an atypical metalloprotease fold that belongs to the gluzincins clan [including, for example, the severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) receptor angiotensin-converting enzyme 2 (ACE2)]. KIAA0895L had escaped annotation as a protease owing to a degenerate signature motif (HExxxH instead of HExxH) that interrupts the active site–containing helix without substantially affecting the integrity of the catalytic site arrangement. We renamed the protein MATCAP (for microtubule-associated tyrosine carboxypeptidase). MATCAP and the vasohibins have adopted fundamentally different catalytic strategies (metalloprotease versus cysteine protease) to target the α-tubulin tail. In addition, whereas vasohibins cleavage relies on the second-last glutamate of the EEY terminal motif and only tolerates the chemically related phenylalanine in place of the α-tubulin terminal tyrosine, MATCAP is promiscuous as to the nature of the terminal residue and shows a dependence on α-tubulin tail residues also located further upstream for substrate recognition. A cryo-EM structure of catalytically inactive MATCAP bound to fully tyrosinated microtubules indicated that MATCAP recognizes tubulin dimers in the same protofilament, whereas the VASH1-SVBP complex sits in between neighboring microtubule protofilaments. Unexpectedly, genetic ablation of MATCAP together with SVBP in mice led to viable offspring with undetectable levels of detyrosination. Despite their viability, these mice displayed morphological brain defects accompanied by proliferation defects during neurogenesis and displayed abnormal behavior.